Zero-shot forecasting: TimesFM-3.0 vs Chronos-2 vs Toto-2.0

18 GIFT-Eval datasets (hourly to monthly, 7,300+ univariate series). GIFT-Eval protocol: horizon 48 (H), 30 (D), 8 (W), 12 (M); rolling non-overlapping test windows over the last 10% of each series (max 20). Multivariate sets are split into univariate series. Up to 600 test windows per dataset (fixed seed). Context = last 1,024 points. Models: Chronos-2 (120M), TimesFM-3.0 (331M), Toto-2.0-313m, CPU, zero-shot. Scores are divided by Seasonal Naive per dataset, then combined with a geometric mean (lower is better; 1.0 = baseline). CRPS = mean weighted quantile loss over the 0.1-0.9 deciles; MASE uses the median.

Key takeaways

• **TimesFM-3.0 is the most accurate on raw data**: relative MASE 0.629 / CRPS 0.470, vs Toto-2.0 0.648 / 0.486 and Chronos-2 0.654 / 0.504. It has the best CRPS on 12 of 18 datasets (Chronos-2 4, Toto-2.0 2; Toto wins solar and hospital). All three beat Seasonal Naive on every dataset by roughly 50% in CRPS, so the gaps between models (3-7%) are small next to their gap to the baseline.
• **Intervals:** TimesFM-3.0 (79.8%) and Toto-2.0 (79.5%) cover close to the nominal 80%; Chronos-2 is slightly too narrow (76.4%). Toto-2.0 never outputs crossed quantiles; TimesFM-3.0 does on 8.5% of forecast steps and Chronos-2 on 3.8%, but sorting them changes CRPS by under 0.1%.
• **Missing values are Chronos-2's weak spot.** On bitbrains (13% missing) its native NaN handling gives 0.94x baseline CRPS; linearly imputing the context first cuts CRPS by 30%. TimesFM-3.0 already interpolates internally and Toto-2.0 masks missing points, so imputation barely moves them, except that back-filling *leading* gaps hurt TimesFM on electricity (+14%), since it normally drops them. Impute interior gaps only.
• **A blanket log1p transform is harmful for every model**: exponentiating the forecast back amplifies the upper tail. TimesFM-3.0 on covid_deaths and Chronos-2 on bitbrains blow up by >1000%, and Toto-2.0 overflows to infinity on one sparse car_parts series. Capping at 10x the history maximum contains the damage (Chronos +1.7%, Toto +5%, TimesFM +16% CRPS) but still does not beat raw. The models scale inputs internally; leave them alone.
• **Outlier clipping (Hampel) hurts all three by 5-7% CRPS.** It removes real signal: with the daily seasonal period of 1 used by GIFT-Eval, it erased the weekly pattern in us_births (+120-170%). It helped only on spiky bitbrains (Chronos -8%, Toto -5%).
• **Context length is the biggest lever.** Cutting context from 1,024 to 256 costs about +10% CRPS for every model (up to +100% on bizitobs and +30-50% on us_births). Going to 2,048 gives a further ~1% (largest for Chronos-2 and Toto-2.0). Give the models as much history as you can afford.
• **Cheap post-processing is safe but nearly free of benefit.** Clipping at zero and rounding count data improve CRPS by under 0.3%. Sign-flip averaging (forecast x and -x, average) helps Chronos-2 and TimesFM-3.0 by ~0.6% at 2x compute and is neutral for Toto. Rescaling interval width using one validation window made things *worse* (+0.7-1.4%; Toto on m4_hourly +17%): the raw intervals are already well calibrated and one window is too noisy to tune on.
• **Overall sensitivity:** Toto-2.0 is the most robust to sensible preprocessing (its worst case is the log transform), Chronos-2 is the most sensitive to how missing data is presented, and TimesFM-3.0 is the most sensitive to the log transform. All three models share the same three harmful steps: short context, log transform, outlier clipping.
• **Caveats:** up to 600 sampled windows per dataset, one seed, univariate mode only, CPU inference (Chronos-2 40 ms, Toto-2.0 49 ms, TimesFM-3.0 53 ms per series). Differences under ~1-2% should be read as ties. TimesFM-3.0 weights are licensed for non-commercial use only.

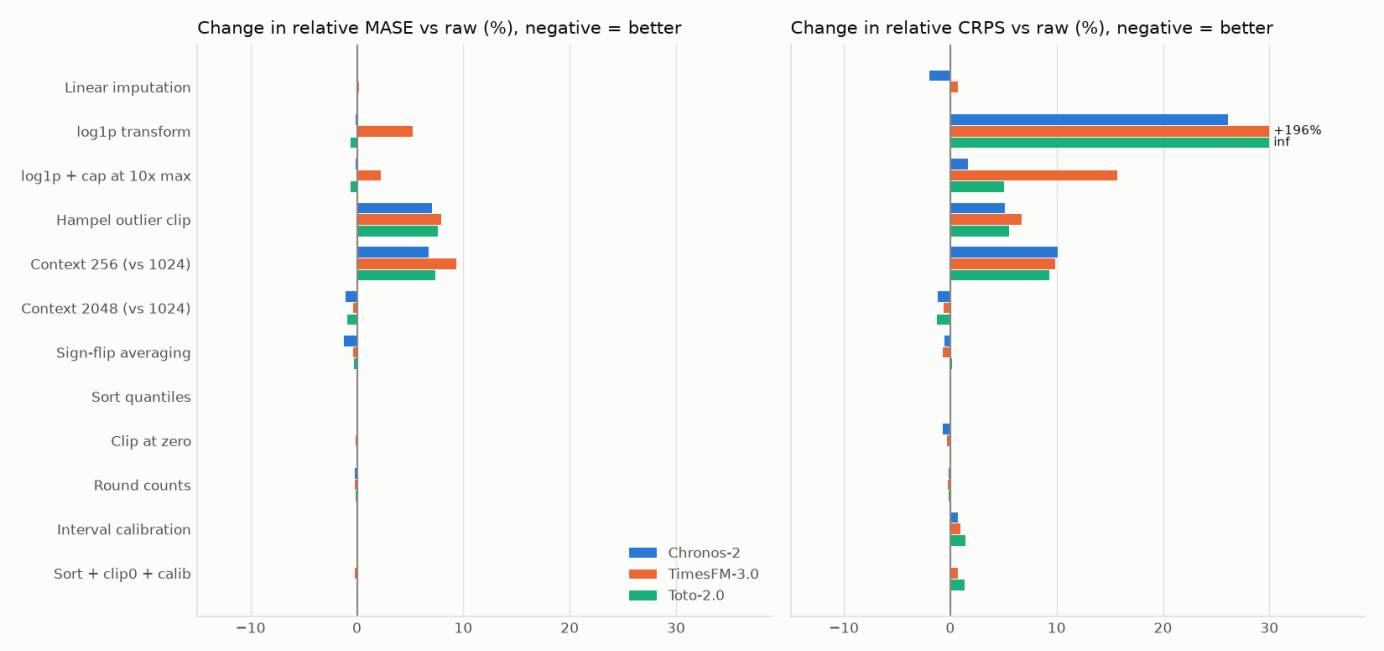
Raw results (lower is better; Seasonal Naive = 1.000)

Model	Rel. MASE	Rel. CRPS	80% coverage	Crossed quantiles	CPU ms / series
Chronos-2	0.654	0.504	76.4%	3.8%	40
TimesFM-3.0	0.629	0.470	79.8%	8.5%	53
Toto-2.0	0.648	0.486	79.5%	0.0%	49

Sensitivity: change vs raw in relative score (negative = better)

Technique	Type	Chronos-2 MASE	TimesFM-3.0 MASE	Toto-2.0 MASE	Chronos-2 CRPS	TimesFM-3.0 CRPS	Toto-2.0 CRPS
Linear imputation	pre	0.0%	+0.2%	0.0%	-2.0%	+0.7%	0.0%
log1p transform	pre	-0.1%	+5.2%	-0.6%	+26.1%	+196.1%	inf
log1p + cap at 10x max	pre+post	-0.1%	+2.3%	-0.6%	+1.7%	+15.7%	+5.0%
Hampel outlier clip	pre	+7.1%	+7.9%	+7.6%	+5.2%	+6.7%	+5.5%
Context 256 (vs 1024)	pre	+6.7%	+9.4%	+7.3%	+10.1%	+9.9%	+9.3%

Context 2048 (vs 1024)	pre	-1.1%	-0.3%	-0.9%	-1.2%	-0.6%	-1.2%
Sign-flip averaging	pre+post	-1.2%	-0.3%	-0.2%	-0.6%	-0.7%	+0.1%
Sort quantiles	post	0.0%	0.0%	0.0%	0.0%	-0.1%	0.0%
Clip at zero	post	0.0%	-0.2%	-0.1%	-0.7%	-0.3%	-0.1%
Round counts	post	-0.2%	-0.2%	-0.1%	-0.2%	-0.2%	-0.1%
Interval calibration	post	0.0%	0.0%	0.0%	+0.7%	+0.9%	+1.4%
Sort + clip0 + calib	post	0.0%	-0.2%	-0.1%	-0.1%	+0.7%	+1.4%



Separate figures in results/figures: dataset_forecasts.png (each dataset with each model's 10-90% prediction range), raw_relative_scores.png, raw_per_dataset_heatmap.png, sensitivity_summary.png, sensitivity_per_dataset.png, interval_coverage.png. Full numbers: results/summary.csv, results/relative_per_dataset.csv.